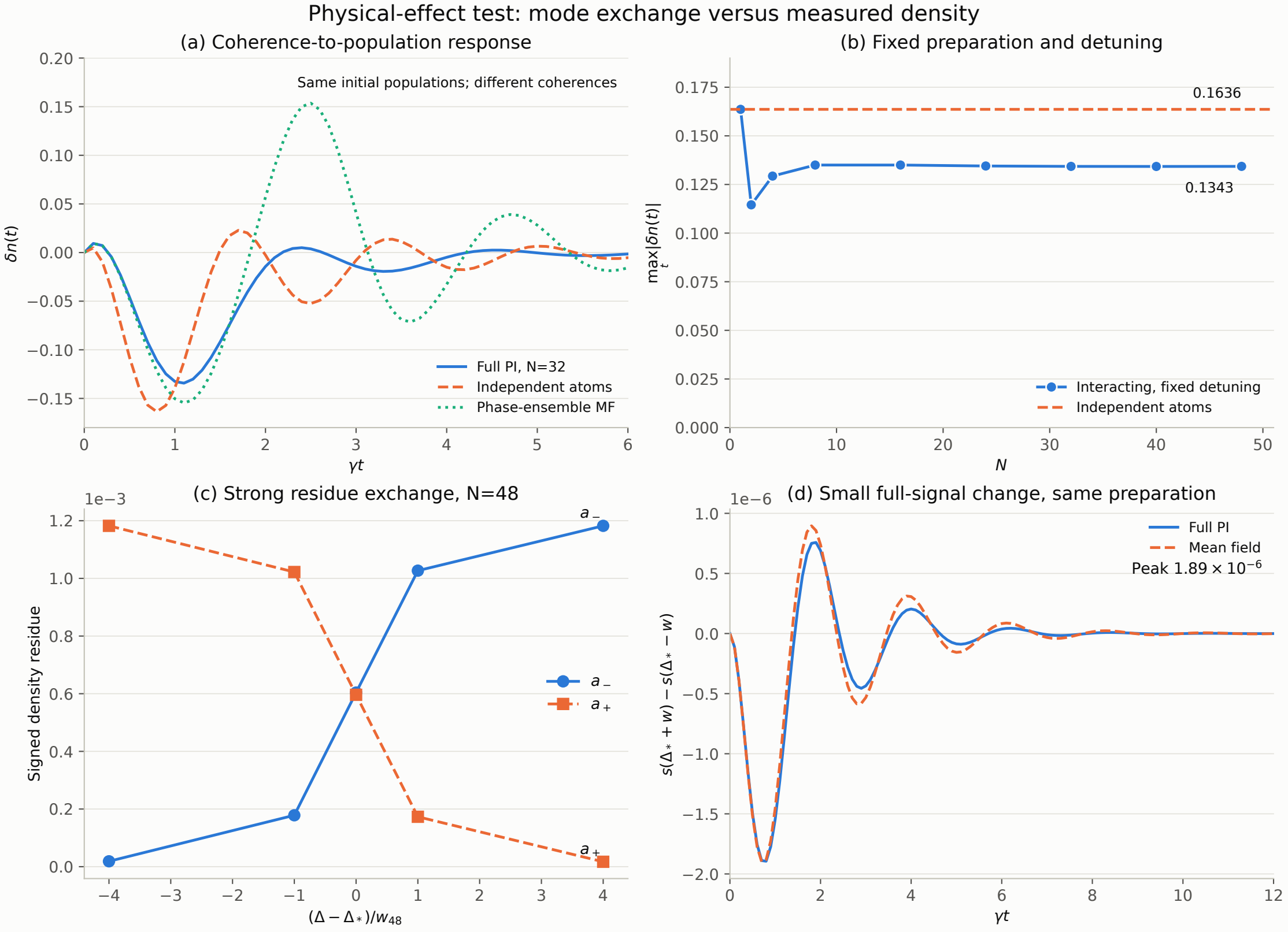
Physical-effect test: mode exchange versus measured density
(a) Coherence-to-population response
Same initial populations; different coherences
Full PI, N=32
Independent atoms
Phase-ensemble MF
δn(t)
γt
(b) Fixed preparation and detuning
0.1636
0.1343
Interacting, fixed detuning
Independent atoms
max_t |δn(t)|
N
(c) Strong residue exchange, N=48
1e−3
a_−
a_+
Signed density residue
(Δ − Δ*)/w48
(d) Small full-signal change, same preparation
1e−6
Full PI
Mean field
Peak 1.89 × 10^−6
s(Δ* + w) − s(Δ* − w)
γt